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MILITARY SPECIFICATION

WASHER, PIPET, AUTOMATIC, SIPHON TUBE TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a siphon tube type plastic pipet washer (rinser) for use on a sink drainboard in washing pipets up to 14 inches long.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-390	- Plastic Molding and Extrusion Material, Polyethylene and Copolymers (Low, Medium and High Density).
L-P-394	- Plastic Molding Material, Polypropylene, Injection and Extrusion.
PPP-B-636	- Boxes, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Headquarters, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, CODE ATT, 2800 South 20th Street, Philadelphia, PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6640

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STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Polyethylene. Polyethylene material shall be in accordance with type I, class L, grade 2 of L-P-390.

3.1.2 Polypropylene. Polypropylene material shall be in accordance with type I of L-P-394.

3.2 Design and construction.

* 3.2.1 Design. The washer shall consist of a cylindrical chamber with a flanged base and siphon tubing attached on the outside of the chamber to connect the bottom of the chamber with the outlet port. The washer shall be designed so that when the inlet port is connected to a water line, the chamber fills and flushes out rapidly through the outlet port. The cylinder chamber shall fill from a level above the siphon tubing (see 6.1) and shall submerge a load of pipets in a pipet basket that has been inserted in the cylinder chamber (see 6.1). When the water level reaches the top of the siphon tubing, a rapid surge of water through the pipet load shall result. The filling and discharging of water in the cylinder chamber shall be continuous and automatic as long as the water line is functioning properly and is connected to the washer.

* 3.2.2 Construction. The walls and bottom of the cylinder chamber and the siphon tubing shall be fabricated from polyethylene not less than 3/64 inch (in) thick. The cylinder chamber shall have a minimum diameter of 6 inches. The chamber shall be sufficiently circular in cross section, and open at the top to allow insertion of a pipet basket having an overall diameter of 5-3/4 in.(see 6.1). The bottom of the cylinder chamber shall have sufficient leveled supports to allow a pipet basket, having a diameter of 4-1/2 in., to stay in a vertical position without toppling against the walls of the cylinder chamber. The siphon tubing shall be permanently attached to the outside of the cylinder. The height of the siphon tubing shall allow the water to fill the chamber at least 17 in. above the

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supports. The outlet port shall be fabricated of polyethylene or polypropylene, and shall be connected permanently to the discharge member of the siphon tubing. The outlet port shall project beyond the outside wall of the cylinder at a level below the top of the supports for a pipet basket. The outlet port shall have a bulb or ridges to accept plastic tubing specified in 3.7. The inlet port shall be fabricated of polyethylene or polypropylene and shall have a bulb or ridges to accept plastic tubing specified in 3.7. The inlet port shall be welded directly to a cutout in the cylinder above the siphon tubing, or it shall be part of a 1/16 inch thick (minimum) polyethylene tube permanently attached to the outside of the cylinder. Inlet tube shall direct the flow of water into the chamber at a level above the top of the siphon tubing. The bottom of the washer shall have a flanged base suitable for supporting the cylinder chamber in an upright, stable position when the washer is placed on a flat, level surface. Slight rocking of the washer, not exceeding 1/16 inch in any direction at base, shall be acceptable. The overall height of the washer, including projections, shall not exceed 30-1/2 inch. The overall width of the washer, including projections, shall not exceed 10-3/4 inch.

3.3 Performance. The washer shall cycle continuously and automatically after connection to a properly adjusted water line. When tested in accordance with 4.4.1, the filling and draining cycle of the washer shall be in accordance with table I; there shall be no evidence of leakage through the walls or joints of the washer components.

TABLE I. Washer, filling and draining cycle.

Inlet flow rate (liters/minute)	Cycle time
2	5 - 6 minutes
12	80 - 110 seconds

3.4 Finish. The manufactuer's commercial colors for the plastic components of the washer shall be acceptable.

3.5 Identification marking. The washer shall be permanently and legibly marked with the name or registered trademark of the manufacturer.

3.6 Workmanship. Workmanship shall be first class throughout. The washer shall be free from defects which detract from its appearance or impair its serviceability. All components of the washer assembly shall be free of deformation, cuts, cracks, chips or exposed sharp edges. The exposed surfaces of the washer shall be essentially smooth. Mold lines shall not be cause for rejection, provided there are no flash protrusions more than 1/16 inch from the surrounding area. All welds that may be applied shall be sound, and they shall be smoothly finished on exposed surfaces. Roughness on the edges of the openings to the top of the cylinder and inlet and outlet ports shall be acceptable, provided there is no evidence of sharpness or projections.

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3.7 Accessories. Flexible plastic tubing shall be supplied (see 3.8). One length of tubing shall be suitable for attachment to the inlet port, and another length of tubing shall be suitable for attachment to the outlet port as a drain extension. The tubing for the inlet port shall have a minimum length of 3-1/2 feet and a nominal inside diameter of 1/2 inch or larger. The tubing for the outlet port shall have a minimum length of 1 foot and a nominal inside diameter of 1/2 inch or larger. The flexible tubing supplied shall be free of cuts, cracks, tears or other defects which impairs its serviceability.

3.8 Service data. Instructions shall be supplied with each washer to indicate the diameters of the flexible tubing (see 3.7) to be used for attachment to the inlet and outlet ports.

4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of three years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the polyethylene or polypropylene, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, the name of the manufacturer or supplier, the NSN, the item identification, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and table II. The inspection levels and acceptable quality levels shall be as indicated in table II. Unit of product for examination purposes shall be the washer.

TABLE II. Sampling for examination.

	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-1	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and table III, except that the acceptance number shall be zero for all sample sizes.

TABLE III. Sampling for tests.

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level
Washer	Performance	3.3	4.4.1	S-1

4.3 Examination. The washer shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. The washer shall be examined for defects including but not limited to those listed in table IV.

TABLE IV. Classification of defects.

Categories	Defects
Major	
101	Components not permanently or securely attached to washer.
102	Components of washer assembly not free of deformation, cuts, cracks, chips or exposed sharp edges.
103	Washer not stable or not free from rocking exceeding 1/16 inch, when placed on a flat and level surface.
104	Components not properly aligned with respect to base of washer.
105	Plastic components not free of imbedded foreign matter.
Minor	
201	Identification marking not legible or complete.
202	Washer not free of excessive flash (i.e., over 1/64 inch).
203	Washer not free of removable dirt, oil or other foreign matter.
204	Plastic components not free of discoloration from normal commercial finish.

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* 4.3.2 Dimensional examination. The washer shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be considered as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

* 4.4.1 Performance test. The inlet of the washer shall be connected to a water supply with a length of flexible tubing conforming to 3.7. A pipet basket shall be placed in the cylinder chamber in such a manner that the handle of the basket does not obstruct the flow of water into the cylinder. The water supply shall be adjusted for flow rates of 12 liters/minute and 2 liters/minute. The rinser system shall operate for 20 minutes at each inlet flow rate, and the filling and draining cycle shall be timed and recorded, using a suitable timing device.

* 4.5 Packaging inspection. The inspection of the unit package, packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

* 5. PACKAGING

5.1 Packaging. Packaging shall be level A or C, as specified (see 6.2).

5.1.1 Unit of issue. One washer constitutes a unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Each washer along with accessories shall be packaged in a box conforming to PPP-B-636, class domestic and having a minimum bursting strength of 175 pounds. The box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to prevent damage to the contents. Closure shall be as specified in the appendix of the box specification.

5.1.3 Level C. Each washer along with accessories shall be packaged in a manner that will afford adequate protection against deterioration and physical damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. Six unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class weather-resistant. Closure, strapping and waterproofing shall be as specified in the appendix of the box specification.

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5.2.2 Level B. Six unit packages shall be packed in a shipping container designed for a type 2 load and conforming to PPP-B-636, class domestic. Closure shall be as specified in the appendix of the box specification.

5.2.3 Level C. The packaged washers shall be packed in shipping containers in a manner that will afford adequate protection against damage during shipment from the supply source to the first receiving activity. These packs shall conform to the carriers' rules and regulations.

5.3 Marking. Each unit package and shipping container shall be marked as specified in MIL-STD-129.

6. NOTES

* 6.1 Intended use. The washer is intended to be suitable for use with the following items that are listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-782-6015	BASKET, PIPET, Plastic
6640-00-782-6007	JAR, PIPET, Plastic

The jar is for soaking of pipets contained in the inserted pipet basket, before the basket is removed from the jar and placed in the washer for final cleaning. A top located water entry is specified in order to preclude contaminated water being siphoned back into the water line.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) National Stock Number (NSN).
(c) Selection of applicable levels of packaging and packing. (see 5.1 and 5.2).

6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet military requirements.

* 6.4 This specification covers the following item listed in the Federal Supply Catalog:

National Stock Number	Item Identification
6640-00-926-7673	WASHER, PIPET, AUTOMATIC, Siphon Tube Type, Single Wall

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6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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MIL-W-36986A WASHER, PIPET, AUTOMATIC, SIPHON TUBE TEST

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